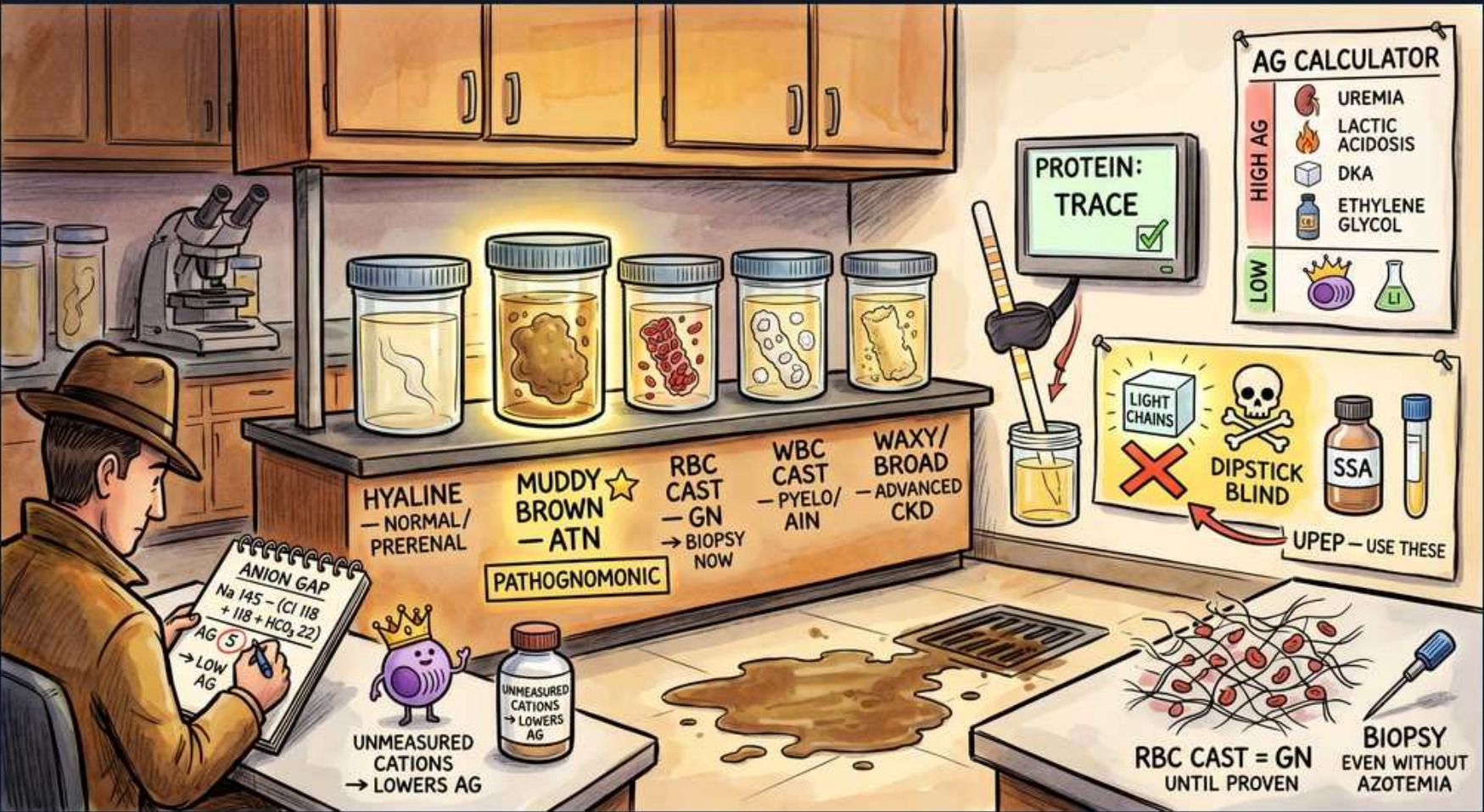


Urine Sediment, Casts & the Anion Gap



1. Hyaline casts → normal/prerenal

WHAT YOU SEE

First jar 'HYALINE — normal/prerenal'

WHAT IT ENCODES

Hyaline casts are acellular Tamm-Horsfall protein seen in concentrated urine, volume depletion, exercise, or diuretic use. They are non-specific and support a bland/prerenal picture rather than intrinsic injury.

BOARD TRAP

Reading hyaline casts as evidence of renal disease — they're benign and favor prerenal physiology.

2. Muddy brown casts → ATN (pathognomonic)

WHAT YOU SEE

Starred jar 'MUDDY BROWN — ATN, PATHOGNOMONIC'

WHAT IT ENCODES

Pigmented muddy-brown granular casts plus renal tubular epithelial cells are essentially pathognomonic for acute tubular necrosis. Their presence converts the diagnosis to intrinsic AKI regardless of FeNa.

BOARD TRAP

Calling muddy brown casts 'nonspecific granular casts' — they specifically indicate tubular necrosis (ATN).

3. RBC casts → GN (biopsy)

WHAT YOU SEE

Jar 'RBC CAST — GN, biopsy now'

WHAT IT ENCODES

RBC casts and dysmorphic RBCs are diagnostic of glomerulonephritis/vasculitis. They mandate serologic workup (ANCA, anti-GBM, complement, ANA) and usually renal biopsy to define the lesion and guide immunosuppression.

BOARD TRAP

Attributing RBC casts to a UTI or stone — they are glomerular until proven otherwise and require biopsy/serologies.

4. WBC casts → pyelo / AIN

WHAT YOU SEE

Jar 'WBC CAST — PYELO/AIN'

WHAT IT ENCODES

WBC casts indicate tubulointerstitial inflammation: acute pyelonephritis or acute interstitial nephritis. With eosinophiluria, fever, rash and a culprit drug (PPI, NSAID, beta-lactam), favor AIN; with bacteriuria/fever, favor pyelonephritis.

BOARD TRAP

Assuming infection only — sterile pyuria with WBC casts and a recent drug points to AIN, not UTI.

5. Waxy/broad casts → advanced CKD

WHAT YOU SEE

Jar 'WAXY/BROAD — advanced CKD'

WHAT IT ENCODES

Broad and waxy casts form in dilated tubules of nephrons with very low flow, marking advanced/chronic kidney disease ('renal failure casts'). They signal chronicity and a poor concentrating reserve.

BOARD TRAP

Interpreting waxy/broad casts as acute injury — they denote chronic, advanced CKD with low tubular flow.

6. Anion gap = $\text{Na} - (\text{Cl} + \text{HCO}_3)$

WHAT YOU SEE

Detective notepad 'AG = $\text{Na} - (\text{Cl} + \text{HCO}_3)$ '

WHAT IT ENCODES

Anion gap = $\text{Na} - (\text{Cl} + \text{HCO}_3)$; normal ~8–12. A high gap means added unmeasured acids; a normal gap means HCO_3 loss with Cl retention. Always check the gap in any metabolic acidosis.

BOARD TRAP

Skipping the gap calculation — a normal HCO_3 with a high gap can hide a mixed disorder.

7. Unmeasured cations lower the AG

WHAT YOU SEE

Crowned ions 'UNMEASURED CATIONS → LOWERS AG'

WHAT IT ENCODES

A low/negative anion gap suggests extra unmeasured cations or loss of unmeasured anions: hypoalbuminemia (commonest — correct AG up ~2.5 per 1 g/dL albumin drop), lithium, multiple myeloma (cationic IgG), bromism.

BOARD TRAP

Interpreting a low gap as 'normal' — hypoalbuminemia masks a true high-gap acidosis; correct the AG for albumin.

8. High AG causes: uremia, lactate, DKA, ethylene glycol

WHAT YOU SEE

'AG CALCULATOR — HIGH: uremia, lactic, DKA, ethylene glycol'

WHAT IT ENCODES

High anion gap acidosis = GOLDMARK/MUDPILES: uremia, lactic acidosis, ketoacidosis (DKA/alcoholic/starvation), toxic alcohols (methanol, ethylene glycol), salicylates. Ethylene glycol also shows an osmolar gap and oxalate crystals.

BOARD TRAP

Forgetting toxic alcohols — a high AG with an elevated osmolar gap and AKI points to ethylene glycol/methanol.

9. Dipstick is blind to light chains

WHAT YOU SEE

'DIPSTICK BLIND — LIGHT CHAINS' skull

WHAT IT ENCODES

The urine dipstick detects albumin only and misses immunoglobulin light chains (Bence-Jones protein). A negative/trace dipstick with heavy proteinuria suspicion should prompt SSA or UPEP to detect non-albumin protein.

BOARD TRAP

Excluding myeloma because the dipstick protein is 'trace' — light chains are invisible to dipstick; order SSA/UPEP.

10. SSA / UPEP detects non-albumin protein

WHAT YOU SEE

'SSA — use these / UPEP' tubes

WHAT IT ENCODES

Sulfosalicylic acid (SSA) precipitates ALL urinary proteins including light chains; a positive SSA with negative dipstick suggests overflow proteinuria from light chains → send UPEP/serum free light chains for myeloma.

BOARD TRAP

Relying on the albumin-specific dipstick alone — SSA/UPEP is required to catch light-chain (overflow) proteinuria.

11. RBC cast = GN until proven otherwise

WHAT YOU SEE

Dysmorphic RBCs 'RBC CAST = GN until proven'

WHAT IT ENCODES

Even without azotemia, dysmorphic RBCs and RBC casts demand a glomerulonephritis workup. Early biopsy can be warranted because crescentic/RPGN disease damages glomeruli rapidly before creatinine rises dramatically.

BOARD TRAP

Waiting for a high creatinine before acting — RBC casts justify urgent GN evaluation regardless of current azotemia.

12. Biopsy even without azotemia

WHAT YOU SEE

'BIOPSY even without azotemia' needle

WHAT IT ENCODES

Glomerular signatures (RBC casts, dysmorphic RBCs, significant proteinuria) justify renal biopsy even when creatinine is normal, since histology guides immunosuppression and prognosis before irreversible scarring.

BOARD TRAP

Deferring biopsy because GFR looks preserved — active glomerular sediment can warrant biopsy with normal creatinine.
